Coherence, resolution and the classical zero-branch limit

For a fixed finite partition with several nondegenerate stationary paths, the coherent pointwise coordinate record converges as its supplied phase-action parameter $h \downarrow 0$ **exactly when every nonzero action-frequency coefficient vanishes** (Theorem 2). Two illuminated branches with distinct actions fail this criterion. Equal actions or several branches permit cancellations; crossing streams alone do not imply failure of a pointwise classical limit.

Smeared records converge to the classical branch-weighted density under the stated nonstationary-phase hypotheses. Independently, Gaussian preparation and coordinate-mark laws for quadratic forces vary continuously with h, and an incoherent mixture of narrowing packets reaches the same classical swarm at every fixed time: weakly as measures, locally uniformly at regular record values (Theorems 1 and 5). Critical values retain their explicit qualifications. These countermodels show why any pointwise interference obstruction requires an actual coherence and resolution premise.

The fringe contrast is dimensionless, not a spectral gap or a selected positive action. The free fold and radial Hooke/Kepler examples retain their exact constructions and local-time hypotheses. Written stationary-phase, Gaussian transport and almost-periodicity arguments, including incorporated internal checks, establish no universal persistence after crossings, necessity of a resolution threshold, or impossibility of classical completion. Model reviews are not independent peer review or novelty evidence.

1. Parameter, preparation and topology

Here h is an arbitrary **reduced phase-action coefficient**, not $h_P = 2\pi\hbar$. Identifying one member with physical quantum dynamics would supply $h = \hbar$; that calibration is not derived. The recording convention is $\kappa = h/2$. Neither h nor a position resolution is the dimensionless $\epsilon = \hbar/S_0$ or a time mesh.

Let T_h denote a specified dynamics, preparation and record law, with classical T_0 as the comparison. A dilation can relate h values only while rescaling the other dimensional data as prescribed by the fifth-postulate and Rivero constructions. Holding the classical preparation, couplings and record protocol fixed while varying h is not merely changing units; the action ratios and interference change.

The claim being tested is continuity of those particular laws, not existence of a classical theory. The oscillatory factors $e^{iS/h}$ can fail pointwise convergence while smooth-observable and fixed-time classical limits exist. Coherent-state transport [@Hepp1974; metadata], Egorov estimates and Wigner measures [@LionsPaul1993; metadata] concern their respective hypotheses and weaker topologies. Exact-record, smeared-record, all-time and refining-instrument limits must therefore be named separately. The groupoid construction provides another continuous state/observable comparison; its continuity does not prove or refute every stronger record limit.

2. Theorem 1: Galileo's comparison has no gap

Theorem 1 (Gaussian preparations, quadratic forces, Gaussian marks). Let $H = p^2/2m + V$ with $\deg V \le 2$ (inertia, uniform gravity, Hooke's force). For each admissible $h \ge 0$ prepare the body with the Gaussian phase-space law $N(z_0, \Sigma)$, $\Sigma > 0$ fixed; for $h > 0$ this is the Gaussian state with that Wigner function, which exists iff $\det \Sigma \ge h^2/4$. Mark the body at times t_j by Gaussian coordinate copies of widths σ_j, each read or forgotten; the pointers are fresh, independent, minimal-uncertainty and unchirped, with coordinate variance σ_j^2 and momentum variance $h^2/(4\sigma_j^2)$, coupled by the shear $z \mapsto z + q$, $p \mapsto p - \pi$. Then the joint law of all read records and of the final coordinate is, for every admissible h and every finite list of marks, the classical Gaussian process of T_0 with one addition: each mark, read or not, adds to the body momentum an independent centred Gaussian kick of variance $h^2/(4\sigma_j^2)$. The family

is continuous in h and equals T_0 at $h = 0$. (A chirped or non-minimal pointer would give a different channel; the conventions are part of the statement.)

Proof. For $\deg V \le 2$ every Moyal flow is the classical affine symplectic flow, and the Wigner function is transported classically (fifth-postulate note, Theorem A(b)). A coordinate copy is generated by $q\pi$, quadratic in the joint canonical variables, so the same holds for the body–pointer shear. A Gaussian Wigner function of a product state is a positive density, and reading the pointer coordinate is classical conditioning of that density. The pointer's momentum π has variance $h^2/(4\sigma_j^2)$ and is independent of its coordinate for a minimal Gaussian pointer, so the shear transfers to the body a kick of that variance whether or not the coordinate is read; this is Proposition 1 of the parallel-move note for the forgotten case, and the read case adds the classical Gaussian conditioning on $r = q + z$. Composition in time is the classical flow between marks. The variance $h^2/(4\sigma_j^2)$ is continuous in h and zero at $h = 0$. □

So on the comparison as the Planck paper sets it, the record family differs from T_0 only by kicks that vanish continuously with h, for every finite list of marks. The zero branch is reached: there is no gap in Galileo's parabola with statistical preparations. (The theorem is a finite calculation; infinite refinement is the separate result that follows.) The same holds for refinement with records at every inserted time: the fixed-h limit exists iff $\sum_j \sigma_j^{-2} < \infty$ (parallel-move note), the $h = 0$ limit always exists, and the $h \to 0$ limit at any fixed partition is continuous. In this limit the zero theory is the regular side and $h > 0$ the singular one.

3. Theorem 2: determinate preparations with several paths

Take the symmetric discrete action of the 1998-conjecture note, (1) there, on a partition with N cells of lengths τ_j, smooth V, fixed initial point $q_0 = x$, record variable $y = q_N$, and internal vertices $z = (q_1, \dots, q_{N-1}) \in \mathbb{R}^d$, $d = N-1$. With a cutoff $\chi \in C_c^\infty(\mathbb{R}^d)$ and a smooth amplitude O, the halved functional with the record variable kept free is

$$A_h(y) = (2\pi h)^{-d/2} \int \chi(z) O(z)\, e^{iS_\pi(x,z,y)/h}\, dz. \tag{1}$$

Its modulus squared $|A_h(y)|^2$ is the complete coordinate record at the final time; $\int \varphi(y)|A_h(y)|^2 dy$ with $\varphi \in C_c^\infty$ is a smeared record.

Hypothesis (several nondegenerate paths). On an open interval Y of record values, $S_\pi(x, \cdot, y)$ has exactly $K \ge 2$ critical points $z_a(y)$ in $\operatorname{supp}\chi$, all nondegenerate, with $\chi = 1$ near each. By the implicit function theorem they depend smoothly on y. Write

$$S_a(y) = S_\pi(x, z_a(y), y), \qquad c_a(y) = \frac{O(z_a(y))\, e^{i\pi\sigma_a/4}}{|\det S_\pi''(z_a(y))|^{1/2}}, \tag{2}$$

with σ_a the Hessian signature.

Lemma (final momenta are distinct). $p_a(y) := S_a'(y)$ equals $m(y - q_{N-1}^{(a)}(y))/\tau_{N-1} - \frac{1}{2}\tau_{N-1}V'(y)$, and $p_a(y) \ne p_b(y)$ for $a \ne b$.

Proof. The envelope formula gives $S_a'(y) = \partial_y S_\pi$ at the critical point, which is the displayed expression. The discrete Euler–Lagrange equation at vertex j, $m(q_{j+1} - q_j)/\tau_j - m(q_j - q_{j-1})/\tau_{j-1} + \frac{1}{2}(\tau_{j-1} + \tau_j)V'(q_j) = 0$, is linear in q_{j-1} with coefficient $m/\tau_{j-1} \ne 0$. So a critical point is determined by (q_{N-1}, q_N) by backward recursion, two distinct critical points have distinct q_{N-1}, and the momenta differ. □

Theorem 2. Under the hypothesis, uniformly for y in compact subsets of Y,

$$|A_h(y)|^2 = \sum_a |c_a(y)|^2 + \sum_{a<b} 2|c_a(y)c_b(y)| \cos\Big(\frac{S_a(y) - S_b(y)}{h} + \theta_{ab}(y)\Big) + O(h). \tag{3}$$

(i) *Complete record.* Fix $y \in Y$ and, for each value ω of $S_a(y) - S_b(y)$ over ordered pairs, put $B_\omega = \sum_{a,b:\, S_a - S_b = \omega} c_a \bar{c}_b$. The limit of $|A_h(y)|^2$ as $h \downarrow 0$ exists iff $B_\omega = 0$ for every $\omega \ne 0$, and

then equals $B_0 = \sum_s \big|\sum_{a:\,S_a=s} c_a\big|^2$, summed over the distinct action values. For $K = 2$ with $S_1(y) \neq S_2(y)$ the condition is $c_1(y)c_2(y) = 0$. The limit B_0 is Newton's branch-weighted density $D_{0,\pi}(|O|^2) = \sum_a |O(z_a)|^2/|\det S''_\pi(z_a)|$ exactly when the total equal-action cross contribution vanishes: $\sum_s \sum_{a<b:\,S_a=S_b=s} 2\,\mathrm{Re}(c_a\bar{c}_b) = 0$. No shared action among illuminated branches is sufficient, not necessary. For example, equal-action coefficients $c_1 = 1, c_2 = i$ give $B_0 = 2$, the diagonal classical value despite both branches being illuminated. (ii) *Smeared record.* For every $\varphi \in C_c^\infty(Y)$,

$$\int \varphi(y)|A_h(y)|^2 dy \longrightarrow \sum_a \int \varphi(y) \frac{|O(z_a(y))|^2}{|\det S''_\pi(z_a(y))|} dy \qquad (h \downarrow 0). \tag{4}$$

Proof. Equation (3) is the stationary phase expansion of (1) with nondegenerate critical points, (3) of the 1998-conjecture note, whose remainder is uniform in the parameter y on compact sets where the critical points stay nondegenerate (Hörmander's stationary phase theorem with parameters; standard). Expanding the modulus squared gives (3) with $\theta_{ab} = \pi(\sigma_a - \sigma_b)/4 + \arg O_a - \arg O_b$.

(i) Put $s = 1/h$. The cross sum in (3) is an almost periodic function $f(s)$ in Bohr's sense, a finite trigonometric sum. If $|A_h(y)|^2$ converges as $h \downarrow 0$ then $f(s)$ converges as $s \to \infty$. An almost periodic function with a limit at infinity is constant: for any s_0 and $\delta > 0$ the set of δ-almost periods is relatively dense, so there are $T_n \to \infty$ with $|f(s_0 + T_n) - f(s_0)| < \delta$, and the limit forces $f(s_0) = \lim f$. A constant trigonometric sum has zero coefficient at every nonzero frequency: the Bohr mean of the leading terms of $|A_h|^2$ against $e^{-i\omega s}$ is B_ω, which collects both orientations of each unordered pair, and the $O(h)$ remainder does not affect it. This proves the condition, and the value B_0 of the limit when it exists.

(ii) In the cross term with indices $a \neq b$ the phase $\psi_{ab} = S_a - S_b$ has $\psi'_{ab} = p_a - p_b \neq 0$ on Y by the Lemma. One integration by parts, $\int g\, e^{i\psi/h} dy = ih \int (g/\psi')'\, e^{i\psi/h} dy$ for $g \in C_c^\infty(Y)$, bounds the term by $O(h)$. The diagonal terms and the uniform remainder give (4). □

Corollary (finite resolution). Let $\varphi \geq 0$ be smooth with compact support and $\int \varphi = 1$, and let $\varphi_\delta(y) = \delta^{-1}\varphi((y - y_0)/\delta)$ be a record of resolution δ at $y_0 \in Y$. With $\Delta p > 0$ the minimum of $|p_a - p_b|$ over pairs $a \neq b$ on the support,

$$\int \varphi_\delta |A_h|^2 dy = \sum_a \int \varphi_\delta \frac{|O(z_a)|^2}{|\det S''_\pi(z_a)|} dy + O\Big(\frac{h}{\delta\,\Delta p}\Big) + O(h). \tag{4'}$$

Here the windows stay in a fixed compact subset of Y and the branch data are bounded there; the kernel derivative contributes the $1/\delta$ and the amplitude derivatives the $O(h)$. So a record of resolution δ reaches Newton's density if $h/(\delta\Delta p) \to 0$: fringes of spacing $h/\Delta p$ are then unresolved. The condition is sufficient, not necessary: vanishing or cancelling amplitudes defeat the fringes, and so can a zero of the kernel's Fourier transform at the fringe wavenumber, for instance a symmetric two-bump kernel at $\delta = \pi h/(2r\Delta p)$. The complete-record topology of (i) is the order of limits in which $\delta \to 0$ before $h \to 0$; the smeared topology of (ii) is the opposite order.

Proof. In the integration by parts of (ii), the derivative of φ_δ carries a factor δ^{-1} relative to φ_δ itself, and $\int \varphi_\delta = 1$; the remaining terms are as before. □

The exact criterion permits non-convergence, a nonclassical limit or the classical limit, depending on the frequency sums and equal-action cross terms. Two illuminated distinct-action branches force non-convergence at the fixed point. Equal-action branches need not give a wrong limit: their cross terms can cancel. The three-path symmetric example in the 1998-conjecture treatment retains its specified nonzero mirror-path cross term; smearing removes it by the distinct arrival-momentum argument. That example is not a theorem about all crossings.

3b. The invariant discontinuous at zero: fringe visibility

A physical spectral gap belongs to a specified Hamiltonian, sector and limiting order. Positivity at every bare coupling is a separate strengthening, not its definition. The witness here is instead the leading dimensionless fringe contrast in (3), under the displayed coherent two-branch assumptions. Its dependence on record resolution supplies no action unit or spectral gap.

Proposition (leading visibility). Let $y_0 \in Y$ be a record point with two illuminated nondegenerate branches, $K = 2$, and write $\rho_a = |c_a(y_0)|^2 = |O(z_a)|^2/|\det S''_\pi(z_a)|$ for Newton's branch densities there, $\rho_1 + \rho_2 > 0$, $\Delta p = |p_1 - p_2|(y_0)$. (i) As $h \downarrow 0$ the complete record near y_0 oscillates with local fringe spacing $2\pi h/\Delta p + O(h^2)$ and visibility

$$V_h = \frac{\max|A_h|^2 - \min|A_h|^2}{\max|A_h|^2 + \min|A_h|^2} = V_0 + O(h), \qquad V_0 = \frac{2\sqrt{\rho_1\rho_2}}{\rho_1 + \rho_2}, \tag{V1}$$

the maximum and minimum taken over one fringe, with the slow variation of the background separated from the interference. The leading term V_0 is fixed by Newton's branch densities and does not depend on h. At $h = 0$ the record is $\rho_1 + \rho_2$ and the visibility is zero. (ii) If $S_1(y_0) \neq S_2(y_0)$, the limit set of $|A_h(y_0)|^2$ as $h \downarrow 0$ is the interval $(\rho_1 + \rho_2)[1 - V_0, 1 + V_0]$. If $S_1(y_0) = S_2(y_0)$ the limit exists and equals $\rho_1 + \rho_2 + 2\sqrt{\rho_1\rho_2}\cos\theta_{12}$. (iii) For a record of resolution δ with kernel φ, the leading cross amplitude is multiplied by $\hat\varphi(\delta\Delta p/h)$, $\hat\varphi(\xi) = \int \varphi(u)e^{i\xi u}du$:

$$V_{\delta,h} = V_0\,|\hat\varphi(\delta\Delta p/h)| + O(h + \delta + \delta^2 \sup|\psi''|/h), \qquad \psi = S_1 - S_2. \tag{V2}$$

For a Gaussian kernel, with its tails controlled, the factor is $\exp[-(\delta\Delta p/h)^2/2]$: full leading contrast when $\delta\Delta p/h \to 0$, none when $\delta\Delta p/h \to \infty$, partial contrast at finite ratios. Other kernels can have Fourier zeros at finite ratios.

Proof. (i) By (3) with $K = 2$, $|A_h(y)|^2 = \rho_1(y) + \rho_2(y) + 2\sqrt{\rho_1\rho_2}(y)\cos(\psi(y)/h + \theta) + O(h)$ with $\psi' = p_1 - p_2 \neq 0$, uniformly near y_0. Over one fringe the cosine runs through ± 1 while the branch densities change by $O(h)$, which gives the spacing, the extremes $\rho_1 + \rho_2 \pm 2\sqrt{\rho_1\rho_2}$ and (V1). (ii) If $\psi(y_0) \neq 0$ then $\psi(y_0)/h$ is continuous and unbounded in h, so its residue mod 2π takes every value along sequences $h \downarrow 0$ and the cosine takes every value in $[-1, 1]$. If $\psi(y_0) = 0$ the cross term is the constant $2\sqrt{\rho_1\rho_2}\cos\theta_{12}$. (iii) Integrate the cross term against φ_δ with $\psi(y) = \psi(y_0) + \Delta p\,(y - y_0) + O(\delta^2 \sup|\psi''|)$ on the window and the coefficients and phase offset frozen at y_0, which costs $O(\delta)$; the result is $2\sqrt{\rho_1\rho_2}\,\mathrm{Re}[e^{i\psi(y_0)/h + i\theta}\hat\varphi(\delta\Delta p/h)]$ up to the stated error, and the diagonal terms are unchanged. □

With $K \geq 3$ illuminated branches, as inside a fold, each pair carries its own leading contrast $2\sqrt{\rho_a\rho_b}/\sum_c \rho_c$ at its own spacing $2\pi h/|p_a - p_b|$. For inertia with graded velocity (Theorem 3) the branch densities are $a(q_a)^2/|1 + p_0'(q_a)t/m|$; for the Kepler swarm (Theorem 4) they are $a(\phi'_a)^2/|1 + I_0'\Omega' t|_a$. These are pair amplitudes before equal action frequencies are grouped. The actual record is governed by the sums B_ω of Theorem 2: cancellations can remove frequencies. The two-branch local leading contrast is independent of h at fixed classical data under its assumptions, but that is a dimensionless record statement. It neither proves a universal pointwise failure at every crossing nor generates a physical action scale.

3c. Imprecision actions at both ends

The record is one end of the comparison; the preparation is the other. A determinate swarm attached to the $h > 0$ theory as one state (5) has no momentum thickness; a real swarm has one, and so does the same classical swarm attached as a mixture of localized states, whose thickness is of order $\sqrt{h}$. The following proposition says what a thickness does, in the exact free setting of §4; the general case follows by the same envelope identity.

Proposition (preparation thickness). Shift the initial curve by a momentum η, $\psi_0^\eta = a\,e^{i(S_0 + \eta q)/h}$, and average the record over η with a density $w_\Delta(\eta) = \Delta^{-1}w_0(\eta/\Delta)$ of width Δ, w_0 fixed with controlled tails. Let x be a record point with exactly two illuminated nondegenerate

arrivals $q_1 \neq q_2$, uniformly across the averaging range, write $\Delta q_0 = |q_1 - q_2|$ for their launch separation, let ϵ_{amp} bound the variation of the branch coefficients across the range and $B = \sup |\partial_\eta (q_1 - q_2)|$ there. Then the averaged record has leading visibility

$$V_{\Delta,h} = V_0 \, |\hat{w}_0(\Delta \, \Delta q_0/h)| + O(h + \epsilon_{\mathrm{amp}} + \Delta^2 B/h), \qquad \hat{w}_0(\xi) = \int w_0(u) e^{i\xi u} du, \tag{V3}$$

so a Gaussian thickness gives the leading factor $\exp[-(\Delta \, \Delta q_0/h)^2/2]$.

Proof. The arrivals solve $q + (p_0(q) + \eta)t/m = x$ and the stationary actions are $\Phi_a^\eta = m(x - q_a)^2/2t + S_0(q_a) + \eta q_a$. By the envelope identity, $d\Phi_a^\eta/d\eta = q_a(\eta)$, so $\Phi_1^\eta - \Phi_2^\eta = \psi(0) + \eta(q_1 - q_2) + O(\eta^2 B)$. Averaging the cross term of (3) against w_Δ with the coefficients frozen gives $2\sqrt{\rho_1 \rho_2} \, \mathrm{Re}[e^{i\psi(0)/h + i\theta} \hat{w}_\Delta((q_1 - q_2)/h)]$ with $\hat{w}_\Delta(\xi) = \hat{w}_0(\Delta\xi)$, up to the stated errors; the diagonal terms change by $O(\epsilon_{\mathrm{amp}})$. □

With both imprecisions present the leading contrast is $V_0 \, |\hat{\varphi}(\delta\Delta p/h)| \, |\hat{w}_0(\Delta \, \Delta q_0/h)|$, provided the joint phase is linearized across both ranges, which adds the condition $\delta\Delta \sup |\partial_x (q_1 - q_2)| \ll h$ to the two curvature conditions. Each argument is an action divided by h: the record imprecision $\delta \, \Delta p$, position resolution times the arrival momentum difference, and the preparation imprecision $\Delta \, \Delta q_0$, momentum thickness times the launch separation. The full leading contrast appears when both imprecision actions are small compared with h; for Gaussian kernels it vanishes when either ratio grows without bound and is partial at finite ratios; other kernels can remove it at finite ratios. This is a different statement from the uncertainty relation, which bounds imprecisions from below by h; here the comparison with Newton needs imprecisions pushed below h. Newton's construction has exact preparations and exact records, both imprecision actions zero, so along the whole family $h > 0$ it sees the leading contrast V_0, and at $h = 0$ none. The two attachments of the same classical swarm are separated by the thickness ratio: the mixture of localized states has $\Delta \sim \sqrt{h}$, so $\Delta \, \Delta q_0/h \to \infty$ and its Gaussian contrast vanishes, while the state (5) has $\Delta = 0$. The state (5) has a definite current velocity p_0/m at each place, not a sharp quantum momentum, and its finite-h Wigner function is not a zero-thickness positive sheet; the premise that selects it over the mixture is determinacy at each h in the sense of a conditional momentum spread of order h (the sheet separation $2\kappa \, \partial_q \log \rho$ of the score note) rather than $\sqrt{h}$. The classical swarm alone does not make that selection. The exact incoherent-limit theorem in §4c does not rely on extrapolating the linearized visibility formula beyond its amplitude and phase-error regime.

Proposition (coherence length). Attach the same swarm as a mixture over centres c of components $a_\sigma(q - c)e^{iS_0(q)/h}$ with $a_\sigma(u) = (2\pi\sigma^2)^{-1/4} e^{-u^2/(4\sigma^2)}$, the centres weighted by a smooth density w on the swarm, with σ independent of h. At a record point with two arrivals $q_1 \neq q_2$, launch separation Δq_0, the leading cross term is the common fringe $2\sqrt{\rho_1 \rho_2} \cos(\psi(x)/h + \theta)$ multiplied by

$$\int a_\sigma(q_1 - c) \, a_\sigma(q_2 - c) \, w(c) \, dc = e^{-\Delta q_0^2/(8\sigma^2)} \, \bar{w} + O(\sigma |w'|), \tag{V4}$$

$\bar{w}$ the weight at the midpoint, while the diagonal terms are $\rho_a \, w(q_a) + O(\sigma|w'|)$. So the leading contrast is $V_0 \, e^{-\Delta q_0^2/(8\sigma^2)}$ up to the weight ratio: partial, the same for every $h > 0$, and zero at $h = 0$. This is (V3) with the minimal thickness $\Delta = h/(2\sigma)$ of such components, since then $\Delta \, \Delta q_0/h = \Delta q_0/(2\sigma)$ and $\exp[-(\Delta \, \Delta q_0/h)^2/2] = e^{-\Delta q_0^2/(8\sigma^2)}$.

Proof. Every component shares the phase S_0, so its arrivals and stationary actions at x are those of Theorem 3 and its fringe phase $\psi(x)/h$ is common; only the amplitudes $a_\sigma(q_a - c)$ depend on the centre. The components are mixed incoherently, so the record is the w-average of the component records. With $(q_1 - c)^2 + (q_2 - c)^2 = 2(c - m)^2 + \Delta q_0^2/2$, m the midpoint, the Gaussian integral over c gives (V4), and $\int a_\sigma(q_a - c)^2 w(c) dc = w(q_a) + O(\sigma|w'|)$. □

The premise therefore has a sharper form than "one coherent state": an h-independent coherence length σ comparable to the launch separation of the crossing streams. Full contrast

as $\sigma/\Delta q_0 \to \infty$, none when $\sigma \to 0$ with h (the $\sqrt{h}$ mixture), and an h-independent partial contrast, hence the same discontinuity at zero, for any fixed $\sigma > 0$. In the optical language this is mutual coherence between the launch points of the two streams. It is this quantity, not purity of the whole swarm, that the second premise must assert at each h.

Proposition (large-action regime). For free motion, or when forces are scaled with mass so the acceleration and trajectories stay fixed, fix $h > 0$ and the geometry of the comparison, and scale the masses by λ at fixed velocities, fixed record resolution δ and fixed velocity imprecision Δv of the preparation. Then momenta, actions, the arrival momentum difference Δp and the momentum thickness $\Delta = \lambda m \Delta v$ all scale by λ, the arrival points, branch densities and V_0 are unchanged, and the two imprecision actions $\delta\Delta p$ and $\Delta\,\Delta q_0$ grow like λ. For Gaussian kernels the leading contrast is $V_0 \exp[-\lambda^2(\delta\Delta p_1/h)^2/2] \exp[-\lambda^2(m\Delta v\,\Delta q_0/h)^2/2]$, with Δp_1 the difference at $\lambda = 1$, and tends to zero as $\lambda \to \infty$; the record converges to Newton's branch-weighted density by Theorem 2(ii) and (4′). The large-action regime at fixed h is therefore the regime in which Newton's mechanics is recovered, and it is defined by comparison with h: the same family read as $h \to 0$ with exact coherent records can fail to converge when Theorem 2(i) has a nonzero action-frequency coefficient.

Proof. Under the scaling the classical trajectories are unchanged, $S \mapsto \lambda S$, and the arguments of $\hat{\varphi}$ and $\hat{w}_0$ in (V2) and (V3) are multiplied by λ; the branch densities are ratios of launch weights to Jacobians of the unchanged arrival map. □

4. Theorem 3: inertial motion with a graded velocity

Free coherent motion permits the pointwise interference test when its illuminated arrival branches satisfy the frequency criterion. Let $h > 0$, $V = 0$, and prepare

$$\psi_0(q) = a(q)e^{iS_0(q)/h}, \qquad a \in C_c^\infty(\mathbb{R}) \text{ real}, \qquad p_0 = S_0'. \tag{5}$$

The classical swarm assigns velocity $p_0(q)/m$ at q; in the quantum state this is a phase-gradient current, not a sharp conditional canonical momentum. The free propagator is exact, so

$$\psi_t(x) = \Big(\frac{m}{2\pi i h t}\Big)^{1/2} \int a(q)\, e^{i\Phi_x(q)/h} dq, \qquad \Phi_x(q) = \frac{m(x-q)^2}{2t} + S_0(q). \tag{6}$$

Critical points of Φ_x are the arrivals $q + p_0(q)t/m = x$, and $\Phi_x''(q) = m/t + p_0'(q)$. Choose a compact interval I containing $\operatorname{supp} a$ and put $M_- = \inf_I p_0'$ and $M_+ = \sup_I p_0'$. Define the candidate fold-time interval by

$$t_* = \frac{m}{\max(0, -M_-)}, \qquad t^* = \frac{m}{\max(0, -M_+)}, \qquad t_* \le t^* \le \infty, \tag{7}$$

with $m/0 = \infty$: for $t_* < t < t^*$ the derivative $1 + p_0'(q)t/m$ takes both signs on I; outside that interval its sign is fixed there. These extrema alone guarantee neither illumination of a fold nor three arrivals.

Theorem 3. (a) For $0 < t < t_*$ the arrival map $q \mapsto q + p_0(q)t/m$ is strictly increasing on I. A record point has at most one illuminated arrival $q_*(x)$, and where it exists $|\psi_t(x)|^2 \to a(q_*(x))^2/(1+p_0'(q_*(x))t/m)$ pointwise, the classical transported density; the limit is zero outside the transported support: the complete record reaches the zero branch before the fold. The same holds for $t > t^*$ when $t^* < \infty$, with the orientation reversed. (b) The candidate fold interval permits several arrivals but does not force an illuminated three-branch interval. On any record interval where exactly $K \ge 2$ illuminated nondegenerate arrivals exist, Theorem 2 applies with $d = 1$ and the free coefficients $c_a = \sqrt{m/t}\, e^{-i\pi/4} a(q_a) e^{i\pi\sigma_a/4}/|\Phi_x''(q_a)|^{1/2}$: the complete record converges as $h \downarrow 0$ only under the frequency condition of Theorem 2(i), and every smearing in x converges to the classical multi-branch density $\sum_a a(q_a)^2/|1 + p_0'(q_a)t/m|$. (c) At a fold point $x_c(t)$ with an isolated illuminated cubic critical point, $\Phi_x''(q_c) = 0$, $b = \Phi_x'''(q_c) \neq 0$, $a(q_c) \neq 0$, and no other equally singular contribution, $|\psi_t(x_c)|^2 \sim (2\pi m/t)\,\mathrm{Ai}(0)^2 (2/|b|)^{2/3} a(q_c)^2\, h^{-1/3}$.

Proof. (a) For $t < t_*$, $1 + p_0'(q)t/m \geq 1 - t\max(0, -M_-)/m > 0$, so the arrival map is strictly increasing on I; one-dimensional stationary phase in (6) gives $|\psi_t|^2 \to (m/t)\, a^2/|\Phi_x''|$, the stated ratio, and zero outside the transported support by nonstationary phase. For $t > t^*$ the derivative is negative throughout and the map is strictly decreasing on I. (b) On a specified interval of several nondegenerate illuminated arrivals, stationary phase applies to each branch. The final momenta $\partial_x \Phi_x(q_a) = m(x - q_a)/t = p_0(q_a)$ are distinct because the q_a are distinct and $x = q_a + p_0(q_a)t/m$. Theorem 2 applies with the displayed coefficients, whose moduli squared are the classical branch densities. (c) Stationary phase at a cubic point gives $|\int a\, e^{i\Phi/h} dq| \sim 2\pi a(q_c)\mathrm{Ai}(0)(2h/|b|)^{1/3}$ against the prefactor $(m/2\pi h t)^{1/2}$; squaring gives the constant and the power. □

A swarm with affine velocity profile $p_0 = -\beta q$, $\beta > 0$, has $t_* = t^* = m/\beta$: it passes through a focal instant and reaches the classical record at every other fixed time. Non-affine profiles require their actual arrival-map test. The existing example $p_0(q) = m(q^2 - q)/t$ at the indicated time gives $F_t(q) = q^2$: derivative sign change and a fold produce two regular arrivals on a symmetric illuminated support, not three. Thus derivative extrema alone did not justify the former three-stream assertion. At an illuminated nondegenerate fold the spatial classical singularity and the $h^{-1/3}$ quantum density asymptotic in part (c) are different limits. Away from critical values, pointwise convergence is governed by the frequency criterion, with smeared convergence as in Theorem 2.

4b. Theorem 4: central forces on the radial cycle, Hooke against Kepler

Newton's central forces supply the case the thesis was made for. Fix the angular momentum $\ell > 0$. The radial motion is one degree of freedom with action $I = I_r$ and angle ϕ on the circle, Hamiltonian $E(I)$, frequency $\Omega(I) = E'(I)$ and radial period $T = 2\pi/\Omega$. The 1998-conjecture note, (10)–(11) gives, for Hooke's force (Book I, Proposition X), $E = \omega(2I + \ell)$, so $\Omega = 2\omega$ and $\Omega' = 0$, the isochrony; and for Kepler's,

$$E = -\frac{mk^2}{2(I+\ell)^2}, \qquad \Omega = \frac{mk^2}{(I+\ell)^3}, \qquad \Omega' = -\frac{3mk^2}{(I+\ell)^4}, \qquad T = 2\pi\sqrt{\frac{ma^3}{k}}. \tag{8}$$

For Kepler the angle ϕ is the mean anomaly, which Kepler's equation turns into position on the ellipse: it is the record astronomy makes.

Model. Take $H_h = E(-ih\partial_\phi)$ on the circle with the twisted periodicity $\psi(\phi + 2\pi) = e^{2\pi i\alpha}\psi(\phi)$, α the Maslov shift of the radial cycle ($\alpha = \frac{1}{2}$), so that the spectrum is $E(h(n + \alpha))$, the Keller–Maslov values, which for Hooke and Kepler are the exact planar spectra ((9)–(11) of that note). Extend E outside the physical range of I by a real smooth function with polynomially controlled derivatives; for preparations whose actions lie compactly inside the physical range, away from the circular boundary $I = 0$, this changes the evolved state by $O(h^\infty)$. The model stipulates the angle record; agreement of its spectrum with the planar one does not by itself identify its angle density with a physical radial or astronomical measurement, and the physical subset of modes also carries the angular condition $\ell/h = |n_\theta|$. The determinate preparation is a swarm,

$$\psi_0(\phi) = a(\phi)e^{iS_0(\phi)/h}, \qquad a \in C_c^\infty((-\pi, \pi)), \qquad I_0 = S_0', \tag{9}$$

each body at ϕ on the orbit of action $I_0(\phi) > 0$, with $I_0(\operatorname{supp} a)$ compact inside the physical range, action spread $\Delta I = \max I_0 - \min I_0$ on the support and support length $|a| < 2\pi$. Poisson summation over the eigenvalue index gives exactly

$$\psi_t(\phi) = \sum_{k\in\mathbb{Z}} \frac{e^{-2\pi ik\alpha}}{2\pi h} \iint a(\phi')\, e^{i\Phi_k(\phi', I)/h}\, d\phi'\, dI, \qquad \Phi_k = S_0(\phi') + I(\phi - \phi' + 2\pi k) - E(I)t. \tag{10}$$

The critical points of Φ_k are $I = I_0(\phi')$ and $\phi - \phi' + 2\pi k = \Omega(I_0(\phi'))t$. The Hessian determinant is $-[1 + tI_0'(\phi')\Omega'(I_0(\phi'))]$. Choose a connected interval $J = [a_-, b]$ of length $\ell_J = b - a_- < 2\pi$ containing the launch support and staying inside the physical action range. Set

$$A_t(\phi') = \phi' + t\Omega(I_0(\phi')), \qquad J_t = 1 + tI_0'\Omega'(I_0),$$
$$t_*^{-1} = \max_J(0, -I_0'\Omega'(I_0)), \tag{11}$$
$$\Delta\Omega = \Omega(I_0(b)) - \Omega(I_0(a_-)).$$

Here $1/0 = \infty$. Before t_*, A_t is increasing on J, and its lifted image width is **exactly** $W(t) = \ell_J + t\Delta\Omega$. The sign matters: a negative $\Delta\Omega$ compresses this interval. A support with holes or zeros can remove illuminated arrivals even when the geometric image overlaps itself on the circle.

Theorem 4 (the stipulated angle-record model).

(a) *Hooke.* For $E(I) = \omega(2I + \ell)$, the I integral in (10) is a delta function and

$$|\psi_t(\phi)|^2 = |a(\phi - 2\omega t)|^2 \pmod{2\pi}. \tag{12}$$

The rigidly rotating angle density is independent of h at all times.

(b) *Nondegenerate arrivals.* If $J_t > 0$ on J and $W(t) < 2\pi$, a circle record has at most one illuminated arrival. At regular values its pointwise limit is $|a(\phi')|^2/|1 + tI_0'(\phi')\Omega'(I_0(\phi'))|$ when that arrival exists and zero otherwise. If $\Delta\Omega > 0$, geometric wrapping starts at

$$t_{\text{wrap}} = \frac{2\pi - \ell_J}{\Delta\Omega},$$

provided the increasing-map condition persists through that time. In the window $2\pi < W(t) < 4\pi$, the circle overlap has two geometric preimages; where both are illuminated, the two-branch stationary-phase criterion applies. Larger widths can give more arrivals, so "exactly two for every later time" is not asserted. More generally, on any interval with finitely many illuminated nondegenerate arrivals, the frequency criterion of Theorem 2 and its smeared-record conclusion apply, with angle ϕ as record and arrival action $I_a = \partial_\phi \Phi_a$ as its momentum. Distinct arrivals have distinct I_a, but their *stationary phase values* can still coincide at a record point; the equal-action cross-term test remains necessary. Resolution $h/(\delta \min_{a\neq b} |I_a - I_b|) \to 0$ is sufficient on the same regular compact windows, not a universal necessary threshold.

(c) *Kepler signs and times.* For (8),

$$t_*^{-1} = 3mk^2 \max_J \left(0, \frac{I_0'}{(I_0 + \ell)^4}\right).$$

If $I_0' < 0$, then $\Omega' < 0$ gives an increasing **stretching** map, $t_* = \infty$ and $\Delta\Omega > 0$; the preceding t_{wrap} applies. If $I_0' > 0$, the map **compresses** before its first fold and cannot wrap there, because $W(t) < \ell_J < 2\pi$. The former positive-gradient pre-fold lapping and half-turn criterion are withdrawn. After complete orientation reversal, if $J_t < 0$ throughout J, its width is $|\ell_J + t\Delta\Omega|$; for $\Delta\Omega < 0$ it reaches 2π at $(2\pi + \ell_J)/|\Delta\Omega|$, only if that monotone reversed regime actually holds. Mixed-sign/folded regimes need the actual arrival count.

For a narrow action spread $\Delta I \ll I + \ell$, the **absolute** frequency spread is $|\Omega'|\Delta I \simeq 3mk^2\Delta I/(I + \ell)^4$. Thus the full-turn dephasing scale and candidate fold scale remain

$$t_{\text{dephase}} \simeq \frac{2\pi(I + \ell)^4}{3mk^2\Delta I} = \frac{T(I + \ell)}{3\Delta I} = \frac{T^2}{\Delta T}, \qquad t_* \simeq \frac{(I + \ell)^4}{3mk^2 \max_J I_0'} \quad (\max_J I_0' > 0), \tag{13}$$

where $\Delta T \simeq 3T\Delta I/(I+\ell)$. A short-support stretching wrap time has the additional factor $1-\ell_J/(2\pi)$; a monotone orientation-reversed wrap time has $1+\ell_J/(2\pi)$. Equation (13) is not a sign-independent theorem of first record overlap. These are classical times, independent of h.

Proof. For the twisted Fourier modes $e^{i(n+\alpha)\phi}$, the preparation coefficients are $c_n = (2\pi)^{-1}\int a(\phi')e^{i[S_0(\phi')-h(n+\alpha)\phi']/h}d\phi'$. Put $\nu = n+\alpha$ and $I = h\nu$. The evolved sum is $\sum_n G(n+\alpha)$, where

$$G(\nu) = (2\pi)^{-1}\int a(\phi')e^{i[S_0(\phi')+h\nu(\phi-\phi')-E(h\nu)t]/h}d\phi'.$$

It is Schwartz in ν under the stated polynomial derivative bounds: outside the compact range of I_0, integration by parts in ϕ' gives rapid decay. Poisson summation, $\sum_n G(n+\alpha) = \sum_k e^{2\pi ik\alpha}\int G(\nu)e^{-2\pi ik\nu}d\nu$, followed by $k\mapsto -k$ and $d\nu = dI/h$, proves (10). For the affine Hooke energy, the I integral is $\delta(\phi-\phi'+2\pi k-2\omega t)$; only one lift contributes because $\ell_J < 2\pi$. Its remaining factors are pure phases, proving (12).

When J_t has one sign, injectivity on J and the **signed endpoint difference** give $W(t)$, or its absolute value for reversed orientation. Reducing an interval of width less than 2π modulo 2π gives at most one preimage; a width between 2π and 4π gives at most two. Stationary phase uses the Hessian $\left(\begin{smallmatrix} I_0' & -1 \\ -1 & -t\Omega' \end{smallmatrix}\right)$, whose determinant is $-J_t$, yielding the stated branch density. If two arrivals have the same I, the arrival equations give $\phi_1'-\phi_2' = 2\pi(k_1-k_2)$; since $\ell_J < 2\pi$ they coincide. Thus distinct arrivals have distinct record momenta, justifying smeared nonstationary phase. Pointwise interference still uses the stationary-action frequency sums. Inserting $\Omega' = -3mk^2/(I+\ell)^4$ proves the sign claims; expanding the frequency difference and $T\propto (I+\ell)^3$ gives (13). □

The observable here is a **stipulated angle record**. Its spectral agreement with the planar problem does not identify it with the physical radial or astronomical measurement. The planar angular condition $\ell/h\in\mathbb{Z}$ also restricts the admitted parameter subsequences. Continuous-h non-convergence need not survive an arbitrary selected subsequence, so the formal angle-family result is not automatically a theorem for that physical slice. A uniform ring at one action is stationary and has no such localization mechanism; a localized coherent swarm needs its actual arrival and frequency conditions. No universal late-time or force-law-only exclusion follows.

For the laboratory coordinate of a harmonic swarm the classical arrival map is $x = q\cos\omega t + p_0(q)\sin\omega t/(m\omega)$, with a fold only where $\cos\omega t + p_0'\sin\omega t/(m\omega) = 0$, and the required illumination and nondegeneracy. This is a different record from the angle model. For a narrow, approximately minimum-uncertainty unchirped angular packet with fixed width σ_ϕ, the estimate $\Delta I \sim h/(2\sigma_\phi)$ makes the dephasing time grow as $1/h$. It gives no universal free-fold time for arbitrary packets or chirps. Fixed-time localization limits require their own shrinking phase-space-width assumptions, as in Theorem 5 below.

4c. Theorem 5: the incoherent attachment reaches zero

The coherence-length proposition of §3c says what a fixed coherence length does. This section removes the premise entirely and proves that without it the gap is absent: a determinate swarm attached to the $h>0$ theory as an incoherent mixture of coherent states whose widths vanish with h has a coordinate law converging weakly to the classical pushforward at every fixed time, with pointwise convergence at regular record values. Critical values retain part (b)'s separate qualifications. The theorem is exact for quadratic forces, the setting of Theorem 3, where a coherent attachment (5) can fail the frequency criterion on a regular multiple-arrival interval. The gap of this note therefore requires the coherence premise.

Setting. Let $H = p^2/2m+V$ with $\deg V\le 2$, so the classical flow Φ_t is affine with linear part $M_t = \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22}\end{pmatrix}$ independent of the phase point. The swarm is the curve $\Lambda = \{(c, p_0(c))\}$,

$p_0 = S_0'$ smooth, carrying the launch density $w \in C_c(\mathbb{R})$, $w \geq 0$, $\int w = 1$; its classical measure is $\mu_\Lambda = w(c)\,\delta(p - p_0(c))\,dc\,dp$. Its arrival map at time t is

$$\begin{aligned} &F_t(c) = \pi_x \Phi_t(c, p_0(c)), \\ &F_t(c) = c + \frac{p_0(c)\,t}{m} \ \text{(inertia)}, \qquad F_t(c) = c\cos\omega t + \frac{p_0(c)\sin\omega t}{m\omega} \ \text{(Hooke)}. \end{aligned} \tag{14}$$

Newton's record of the swarm is the pushforward $(F_t)_*(w\,dc)$: at a regular value x of F_t, with preimages q_a in the support and $F_t'(q_a) \neq 0$, it has the density $\sum_a w(q_a)/|F_t'(q_a)|$; at an isolated nondegenerate fold with positive launch weight it has a one-sided inverse-square-root singularity, and at other critical values its behaviour depends on the degeneracy and on the weight.

For $h > 0$ attach the swarm as the mixture

$$\varrho_h = \int w(c)\,|\varphi_c^h\rangle\langle\varphi_c^h|\,dc, \qquad \varphi_c^h(q) = (2\pi\sigma_h^2)^{-1/4} \exp\Big[-\frac{(q-c)^2}{4\sigma_h^2} + \frac{i\,p_0(c)(q-c)}{h}\Big], \tag{15}$$

coherent states in the sense of minimum-uncertainty Gaussian packets, squeezed relative to any fixed oscillator, centred at the swarm's phase points with position width σ_h and momentum width $h/(2\sigma_h)$, both vanishing with h: $\sigma_h \to 0$ and $h/\sigma_h \to 0$, for instance $\sigma_h^2 = hL_*/P_*$ with a fixed length L_* and a fixed momentum P_*. The Wigner function of φ_c^h is the Gaussian with mean $(c, p_0(c))$ and covariance $\Sigma_h = \mathrm{diag}(\sigma_h^2,\, h^2/4\sigma_h^2)$; the Wigner function of ϱ_h is its w-average, a positive density that converges weakly to μ_Λ. So (15) is the same determinate swarm, attached without coherence between centres.

Theorem 5 (coherence necessity). Under the setting above, at every fixed t put $s_t(h)^2 = M_{11}^2\sigma_h^2 + M_{12}^2 h^2/(4\sigma_h^2)$; then $s_t(h) > 0$, since the first row of the symplectic M_t cannot vanish, and $s_t(h) \to 0$ as $h \to 0$ at each fixed t. Then: (a) the complete coordinate record of (15) is exactly

$$\rho_h(x,t) = \int w(c)\,N\big(x;\,F_t(c),\,s_t(h)^2\big)\,dc, \tag{16}$$

a positive density with no oscillatory term; (b) $\rho_h(\cdot,t) \to (F_t)_*(w\,dc)$ weakly as $h \to 0$, and at every regular value x of F_t, with w continuous,

$$\rho_h(x,t) \longrightarrow \sum_a \frac{w(q_a)}{|F_t'(q_a)|} \qquad (h \to 0), \tag{17}$$

Newton's branch-weighted density, the convergence being uniform on compact sets of regular values; at a critical value the weak convergence still holds, while the pointwise behaviour depends on the degeneracy and the launch weight: at an isolated nondegenerate fold $F_t''(q_c) \neq 0$ with $w(q_c) > 0$ the local contribution to ρ_h scales as a positive constant times $s_t(h)^{-1/2}$ and the classical density has a one-sided $|x - x_c|^{-1/2}$ singularity, whereas a weight vanishing at the fold can make both finite; (c) in the setting of Theorem 3, inertia with $w = a^2$, $\int a^2 = 1$, on any regular record interval inside the fold interval carrying exactly $K \geq 2$ illuminated arrivals, the incoherent attachment converges to $\sum_a a(q_a)^2/|1 + p_0'(q_a)t/m|$, where the coherent attachment (5) converges only under the frequency condition of Theorem 2(i), with the equal-action value B_0 when it does. The two attachments are different states at finite h, with the same classical limit on smeared records by Theorem 2(ii); they differ in the complete record. Their difference in coherence is exact: the initial kernel of (15) obeys $|\varrho_h(q,q')| \leq \|w\|_\infty e^{-(q-q')^2/(8\sigma_h^2)}$, so the coherence between any two fixed distinct launch points vanishes as $\sigma_h \to 0$, while the state (5) has $|\psi_0(q)\psi_0(q')| = |a(q)a(q')|$. The gap of Theorems 2–4 therefore requires coherence between the launch points of the crossing streams.

Proof. (a) For real quadratic V the Moyal corrections to the Wigner equation vanish identically, so every Wigner function obeys $\partial_t W + (p/m)\partial_q W - V'(q)\partial_p W = 0$ and is transported by the classical affine flow, $W_t = W_0 \circ \Phi_{-t}$; this is the content of Theorem A(b) of the fifth-postulate note read on states. Hence the Wigner function of $U_h(t)\varphi_c^h$ is the Gaussian with mean $\Phi_t(c, p_0(c))$

and covariance $M_t\Sigma_h M_t^{\mathsf{T}}$, whose position marginal is the Gaussian $N(x; F_t(c), s_t(h)^2)$ with $s_t(h)^2 = (M_t\Sigma_h M_t^{\mathsf{T}})_{11}$, the displayed expression. Densities of a mixture add with the mixture's weights, which gives (16). Since M_t does not depend on the phase point, $s_t(h)$ does not depend on c, and $s_t(h) \to 0$ because $\sigma_h \to 0$ and $h/\sigma_h \to 0$. (b) *Weak convergence.* For bounded continuous g, $\int g\rho_h\,dx = \int w(c)\big[\int g(x)N(x; F_t(c), s_t^2)dx\big]dc$; the inner integral tends to $g(F_t(c))$ for every c and is bounded by $\sup|g|$, so dominated convergence gives $\int w(c)g(F_t(c))dc = \int g\,d(F_t)_*(w\,dc)$. *Pointwise convergence.* Let x be a regular value and $q_1, \dots, q_K$ its preimages in $\operatorname{supp} w$, finitely many because they are isolated in a compact set. Choose disjoint neighbourhoods U_a of the q_a on which F_t is a diffeomorphism onto a neighbourhood of x, and $\eta > 0$ with $|F_t(c) - x| \geq \eta$ for $c \in \operatorname{supp} w \setminus \bigcup_a U_a$, which exists because that set is compact and contains no preimage. The contribution of that set to (16) is at most $\|w\|_\infty |\operatorname{supp} w|\,(2\pi s_t^2)^{-1/2} e^{-\eta^2/2s_t^2} \to 0$. On U_a substitute $u = F_t(c)$: $\int_{U_a} w(c)N(x; F_t(c), s_t^2)dc = \int w(F_t^{-1}(u))\,N(x; u, s_t^2)\,\frac{du}{|F_t'(F_t^{-1}(u))|} \to \frac{w(q_a)}{|F_t'(q_a)|}$ by continuity of the integrand at $u = x$. Summing over a gives (17). For uniformity on a compact set R of regular values, cover the compact set $F_t^{-1}(R) \cap \operatorname{supp} w$ by finitely many inverse charts of F_t, on which the local densities $w \circ F_t^{-1}/|F_t' \circ F_t^{-1}|$ are uniformly continuous, and choose one $\eta > 0$ separating R from the images of the remaining compact part of the support, which contains no preimage of R; the tail bound is then uniform and the approximate identity converges uniformly on R. At a critical value the substitution fails; the weak convergence of (b) is unaffected. At an isolated nondegenerate fold with $w(q_c) > 0$, $F_t(c) - x_c \simeq \frac{1}{2}F_t''(q_c)(c - q_c)^2$, and the substitution $c - q_c = s_t^{1/2}u$ in (16) gives a local contribution $w(q_c)\,s_t^{-1/2}\int(2\pi)^{-1/2}e^{-F_t''(q_c)^2u^4/8}du + o(s_t^{-1/2})$, while the classical density near x_c is $w(q_c)|2/(F_t''(q_c)(x - x_c))|^{1/2}$ on the covered side. If instead w vanishes at the fold, both can stay finite: for free motion with $p_0(c) = (c^2 - c)m/t$, so that $F_t(c) = c^2$, and $w(c) = Cc^2\chi(c)$ with a bump χ, the classical density near $x_c = 0$ is $C\sqrt{x}$ and $\rho_h(0, t) = O(s_t^{1/2}) \to 0$. (c) For inertia $M_t = \begin{pmatrix} 1 & t/m \\ 0 & 1 \end{pmatrix}$, so $s_t(h)^2 = \sigma_h^2 + t^2h^2/(4m^2\sigma_h^2) \to 0$ and $F_t'(c) = 1 + p_0'(c)t/m$; on a regular interval with K illuminated arrivals, (17) is the stated K-branch density. Theorem 3(b) and Theorem 2(i) give the coherent attachment's behaviour and Theorem 2(ii) the common smeared limit. For the kernel bound, (15) gives $|\varphi_c^h(q)\varphi_c^h(q')| = (2\pi\sigma_h^2)^{-1/2}e^{-[(q-c)^2+(q'-c)^2]/4\sigma_h^2}$, and $(q - c)^2 + (q' - c)^2 = 2(c - \bar{q})^2 + \frac{1}{2}(q - q')^2$ with $\bar{q}$ the midpoint; integrating the Gaussian in c against $w \leq \|w\|_\infty$ gives the bound. □

Scope. The theorem is exact for quadratic forces, which include the free fold of Theorem 3 and Hooke's laboratory record. For a general smooth force the same conclusion, in the weak form of (b), follows at fixed time from the semiclassical propagation of coherent states along classical trajectories [@Hepp1974; metadata] or from the transport of Wigner measures [@LionsPaul1993; metadata], under the hypotheses those theorems place on the potential, the flow and the admissible squeezing of σ_h; Kepler's $1/r$ lies outside the usual symbol classes, and the angle-record model of Theorem 4 with a polynomially controlled E lies inside them. Those extensions are cited, not proved here. What is proved is the statement the thesis needs: in the very setting where the coherent attachment shows the gap, the incoherent attachment of the same determinate swarm does not.

5. All-time invariants

Two quantities already in the corpus show the other non-commuting pair of limits, $h \to 0$ against $t \to \infty$. For the free Gaussian, the expected number of sign reversals of the two-sheet realization over all time is $\pi\varphi(1)$ for every $h > 0$ and zero at $h = 0$ (recording note, (150)); the total phase $\arctan u$ of (143) there reaches $\pi/2$ for every $h > 0$ and is zero at $h = 0$. Both are limits taken at infinite time first; at fixed finite time both vanish continuously as $h \to 0$. The first is a hidden count and the second a phase, so neither is a record; their observable counterpart is the breakdown of the classical limit after the Ehrenfest time, which Theorem 3 exhibits inside the

fold interval for inertial motion. These examples compare orders of limits; without an actual observable they give no additional complete-record theorem or positive-action selection.

6. Consequence for STATE

The pointwise coherent criterion, finite-resolution estimate and incoherent countermodel answer different limit questions. On compact regular record windows, $h/(\delta\Delta p) \to 0$ is a **sufficient** resolution condition for the classical smeared limit, not a necessary one; frequency cancellation and Fourier zeros are retained exceptions. The preparation-thickness and coherence-length propositions give their additional curvature and amplitude conditions. A classical swarm with one velocity at each launch point does not choose a coherent quantum attachment over an incoherent one.

For quadratic dynamics, the Gaussian marks add recoil variance $h^2/(4\sigma_j^2)$ per mark. Thus the finite-record family is continuous in h, not constant in h. Under coherent two-branch hypotheses the leading visibility can instead remain $2\sqrt{\rho_1\rho_2}/(\rho_1+\rho_2)$ as $h \downarrow 0$. These statements coexist because their preparations and record limits differ. The Hooke/Kepler angle examples retain their signed wrapping conditions; $T^2/\Delta T$ is the narrow-swarm full-turn dephasing scale, not a sign-independent first-overlap theorem. There is no universal late-time obstruction.

A vanishing-action limit and a large-action regime at fixed h also vary different data. Their distinction does not prove that classical mechanics exists only as one of them. Classical initial-value construction, cusp-history completion and singular domain selection have separate hypotheses. The interpretation of exact Newtonian geometry or Leibniz's text as a physical complete-record rule remains a historical/physical premise, not a consequence of the interference calculation.

The open selection question is whether an independently justified physical preparation and instrument law enforces the required coherence and resolution, and then whether any resulting coefficient is universal and equals physical $\hbar$. The constructions here supply no such necessity or calibration. This treatment remains separate from the Gaussian mark bound and classical existence proofs for its exact multi-frequency criterion, coherent/incoherent attachments, fold/angle constructions and two-ended imprecision laws. New research remains paused during consolidation.